NAHID HASAN

AI ENGINEER | NLP | MULTIMODAL LEARNING | APPLIED LLM SYSTEMS

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Professional Summary

Computer Science and Engineering graduate with experience in NLP, multimodal deep learning, and retrieval-augmented generation (RAG). Published peer-reviewed Bangla NLP research at BLP-2025 (IJCNLP-AACL). Skilled in Python, PyTorch, TensorFlow, and Hugging Face, with practical experience in AI prototyping, model evaluation, and software development.

Education

Rajshahi University of Engineering & Technology (RUET) Mar 2022 – Aug 2026
B.Sc. in Computer Science & Engineering | **CGPA: 3.68/4.00**
Undergraduate thesis: CAAM-Net: A Conflict-Aware Attention-Based Multimodal Network for Bangla Clickbait Detection

Technical Skills

AI/ML: NLP, Transformers, Multimodal Learning, Deep Learning, Ensemble and Multi-task Learning, RAG, LLM APIs, Embeddings, Vector Search, Model Evaluation
Libraries/Frameworks: PyTorch, TensorFlow, TensorFlow Lite, Hugging Face, scikit-learn, NumPy, Pandas, Flask
Programming/Tools: Python, C, C++, Java, Git, GitHub, Linux, PostgreSQL, pgvector, MySQL, SQLite, Railway

Selected AI Projects

AinBondhu: Source-Grounded Legal Q&A Assistant | GitHub 2026
- Built and deployed a criminal-law RAG assistant with Flask, Gemini embeddings, PostgreSQL and pgvector search, and Groq LLM generation.
- Implemented section/page-grounded citations, version-aware document ingestion, and retrieval evaluation. Live demo.

CAAM-Net: Multimodal Bangla Clickbait Detection | Undergraduate Thesis 2025 – 2026
- Developed a conflict-aware attention-based model that learns relationships between video titles and thumbnails to detect Bangla clickbait.
- Achieved the best overall performance among evaluated baselines, with only a 0.0129 Macro-F1 drop under conflicting title–thumbnail conditions.

Bangla Hate Speech Detection | Transformer Ensembles and Multi-task Learning 2025
- Combined multilingual transformers, ensemble approaches, and multi-task learning to classify hate type, target, and severity.
- Evaluated alternative transformer and multi-task configurations across three subtasks; published findings at BLP-2025.

AI-Powered Plant Disease Detection | Hackathon Project 2025
- Contributed to leaf-image disease detection with bilingual care guidance and integrated a Bangla–English chatbot using RAG, LLMs, and web fallback.
- **Stack:** Flask, TensorFlow Lite, SQLite, RAG | GitHub repository

Industry Experience

Ankabut Software Company Ltd. | Industrial Attachment Trainee Mar 2025
- Contributed to an AI-powered multi-company chatbot for website navigation, information access, and service discovery.
- Participated in requirements analysis, system design, prototype development, testing, and Agile workflows.

Additional Software Project

Profit Ex | Financial Management Web Application 2024
Built a Flask/Python/MySQL web application for income, expense, salary, and profit tracking, with dashboards, PDF reports, and user authentication. GitHub repository.

Publication

Hasan, N. (2025). Catalyst at BLP-2025 Task 1: Transformer Ensembles and Multi-task Learning Approaches for Bangla Hate Speech Detection. *Proceedings of the Second Workshop on Bangla Language Processing*, 468–475. DOI link.

Reference

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